Improved Sentiment Analysis Model: From Data Balancing to Deep Learning Deployment

Team ScourgifyData

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1 Introduction

Following our baseline logistic regression model, we identified class imbalance as the primary limitation. This report documents: (1) data balancing via upsampling, (2) deep learning training using TensorFlow on Kaggle, and (3) model evaluation with accuracy and predictions.

2 Phase 1: Data Balancing Through Upsampling

2.1 Problem Identification

Our dataset exhibited severe class imbalance: positive reviews dominated while negative and neutral were significantly underrepresented. This causes models to bias toward the majority class, resulting in poor minority class performance and missing critical customer feedback.

2.2 Upsampling Methodology

We implemented intelligent upsampling using synonym-based text augmentation with **NLPAug**, **WordNet**, and **NLTK**. Process: (1) separated dataset by sentiment class, (2) identified target size (majority class size), (3) for minority classes, generated synthetic samples by replacing ∼10% of words with synonyms from WordNet, (4) cycled through originals creating variations while maintaining semantic meaning, (5) repeated until all classes equal, (6) combined and shuffled (seed 42).

2.3 Upsampling Results

Table 1: Dataset Distribution Before/After Upsampling

Class	**Before**	**After**	**Increase**
Positive	n	n	0%
Negative	$0.3n$	n	233%
Neutral	$0.15n$	n	567%
Total	$1.45n$	$3n$	107%

Benefits: Equal representation prevents bias, synonym replacement creates realistic variations, linguistic diversity improves generalization, enables equal learning across classes. Balanced dataset saved as `model_balanced.csv`, split 80% train / 20% validation.

3 Phase 2: Model Training with TensorFlow on Kaggle

3.1 Training Environment

We used Kaggle Notebooks for: free GPU acceleration, pre-installed TensorFlow libraries, high computational resources, and easy dataset management. Uploaded `train.csv` (80%) and `val.csv` (20%) as inputs.

3.2 Model Architecture: Bidirectional LSTM

We designed a deep neural network using TensorFlow/Keras to capture sequential patterns and context in review text, overcoming baseline's bag-of-words limitations.

Table 2: Neural Network Architecture

Layer	Config	Purpose
TextVectorization	60K vocab, len=200	Text to integers
Embedding	128D	Dense word vectors
BiLSTM	128 units	Bidirectional context
GlobalMaxPool	–	Extract key features
Dense + Dropout	64 units, 40% drop	Transform & regularize
Output	3 units, Softmax	Class probabilities

Key features: BiLSTM processes text bidirectionally (understands negation), embedding learns semantic similarity, GlobalMaxPooling handles variable lengths, dropout prevents overfitting.

3.3 Training Configuration

Preprocessing: Lowercase conversion, URL/mention/hashtag removal, letter-only retention, whitespace normalization. Labels encoded: negative=0, neutral=1, positive=2.

Parameters: Loss: Sparse Categorical Cross-entropy, Optimizer: Adam (lr=0.001), Batch: 4,096, Epochs: 3, Pipeline: TensorFlow Dataset API with shuffling/batching/prefetching.

Workflow: (1) Load datasets, (2) preprocess & encode, (3) create TF Dataset objects, (4) build BiLSTM architecture, (5) compile model, (6) train 3 epochs with GPU, (7) monitor validation metrics, (8) save as `sentiment_model.keras`.

3.4 Training Results

Model converged successfully: high accuracy on training/validation sets, no significant overfitting, balanced learning across classes, ∼30 min training time with GPU.

4 Phase 3: Model Evaluation and Predictions

4.1 Deployment & Performance

Model saved as `sentiment_model.keras`, production-ready deployment pipeline: loads model, preprocesses text, generates predictions with probabilities, displays results.

Table 3: Model Performance Metrics

Metric	Score
Overall Accuracy	High (88%)
Balanced Performance	Equal across classes
Negative Recall	Significantly improved
Neutral Recall	Significantly improved
Positive Recall	High

Improvements: Dramatic neutral detection boost, better negative identification, enhanced negation understanding, balanced F1-scores, improved probability calibration.

4.2 Prediction Examples

Table 4: Sample Predictions

Review Text	Predicted	Confidence
"Hotel absolutely amazing! Staff friendly, view breathtaking."	Positive	Very High
"Tour okay, nothing special but not terrible."	Neutral	High
"Worst experience. Room dirty, service horrible."	Negative	Very High
"Beautiful beach, perfect weather! Highly recommend!"	Positive	Very High
"Museum interesting, decent collection."	Neutral	Moderate
"Food delicious but restaurant was dirty."	Negative	Moderate

Analysis: Confidently identifies clear positive/negative reviews, successfully recognizes neutral language, handles mixed sentiment cases, captures negation/intensifiers, performs well on tourism vocabulary.

4.3 Baseline Comparison

Table 5: Baseline vs. Improved Model

Aspect	Baseline	Improved
Architecture	LogReg	BiLSTM
Features	TF-IDF (10K)	Embeddings (128D)
Context	No	Yes (bidirectional)
Data	Imbalanced	Balanced
Platform	Local	Kaggle GPU
Time	<1 min	∼30 min
Accuracy	Moderate	High
Minority recall	Poor	Improved
Negation	Limited	Strong

5 Conclusion

This project developed an improved sentiment analysis model through three phases: (1) intelligent upsampling achieving perfect class balance, (2) Bidirectional LSTM with TensorFlow capturing contextual dependencies, (3) Kaggle GPU training and production deployment.

Business Value: Provides tourism businesses accurate sentiment analysis across all feedback types, enabling identification of service issues, satisfied customers, and areas needing attention. Balanced performance ensures no critical feedback is overlooked.